

EEL4712 Digital Design

Lecture 29 – System Verilog Assertions III

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Agenda

- Assertion properties vs cover properties
- Cover properties
- Cover properties in Questa

Assertion Coverage

- Purpose
  - Assertion property verifies correct behavior over time
  - Properties can also be used to measure how often behaviors occur
  - Coverage helps determine whether important scenarios were exercised
- Two main uses of property:
  - Checking behavior → assert property
  - Measuring behavior → cover property

```
assert property (@(posedge clk)
    req |-> ##[1:3] ack
);
```

If req occurs, ack must occur within 1–3 cycles

If violated → **simulation error**

Using Cover Property

- Measuring Scenario Occurrence
- Assertions can also track whether a scenario actually happens.
- Example:

```
cover property (@(posedge clk)
   req ##1 ack);
```

- Meaning:
  - Record coverage if req is followed by ack 1 cycle later

Statement	Purpose
assert property	Detect incorrect behavior
cover property	Measure if behavior occurred

Example: Handshake Coverage

- Design Requirement
- req → ack within 1–3 cycles
- Assertion

```
assert property (@(posedge clk)
  req |-> ##[1:3] ack);
```

- Coverage

```
cover property (@(posedge clk)
  req ##[1:3] ack);
```

- Coverage tells us:
  - Did this handshake ever occur?

Cover Properties in Questa

- Compile, Simulate, and Analyze
  - vlog -sv design.sv tb_design.sv
  - vsim +coverage tb_design
  - run –all
  - `coverage report -assert -details`

In Class Demo – Cover Property DFF

```
// 1. Did we ever reset the FF?
cover property (@(posedge clk) rst);

 // 2. Did we see a load event (en=1 and q updates next cycle)?
 cover property (@(posedge clk)
     en ##1 (out == $past(in))
 );

 // 3. Did we see both enable and disable behavior?
 cover property (@(posedge clk)
     en ##1 !en
 );
```

What does the cover property report tell us?

```
# DIRECTIVE COVERAGE:
# --------------------------------------------------------------------------------------------
# Name                                  Design Design   Lang File(Line)      Hits Status
#                                       Unit   UnitType
# --------------------------------------------------------------------------------------------
# /tb_ff_cover/cover__2                 tb_ff_cover Verilog  SVA  /ecel/UFAD/rewetz/Desktop/assertions/tb_ff_cover.sv(54)
#                                                                                2473 Covered
# /tb_ff_cover/cover__1                 tb_ff_cover Verilog  SVA  /ecel/UFAD/rewetz/Desktop/assertions/tb_ff_cover.sv(49)
#                                                                                5001 Covered
# /tb_ff_cover/cover__0                 tb_ff_cover Verilog  SVA  /ecel/UFAD/rewetz/Desktop/assertions/tb_ff_cover.sv(44)
#                                                                                   0 ZERO
#
# TOTAL DIRECTIVE COVERAGE: 66.66%  COVERS: 3
#
# Total Coverage By Instance (filtered view): 66.66%
```

```
initial begin : drive_inputs
    $timeformat(-9, 0, " ns");

    rst = 1'b0;
    in  = 1'b0;
    en  = 1'b1;

    repeat (5) @(posedge clk);
    @(negedge clk);
    rst = 1'b0;


    for (int i = 0; i < NUM_TESTS; i++) begin
        @(negedge clk)
        in = $urandom;
        en = $urandom;

        @(posedge clk);
    end

    disable generate_clock;
    $display("Tests completed.");
end
```

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```

The first property is not covered. If we look at the testbench, it can be confirmed that the design is never RESET!

```
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    repeat (5) @(posedge clk);
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        @(posedge clk);
    end

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